



RADHIKA DOTIHAL

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Software Engineer experienced in developing and supporting scalable applications using C++, Java, Python, MySQL and Podman. Adept at finding efficient solutions to complex challenges, writing efficient and maintainable code, collaborating with cross-functional teams. I thrive on continuous learning to create impactful, revenue boosting products.

PROFESSIONAL EXPERIENCE

Software Engineer –SS&C Technologies, Inc. Advent [C++, Python, MySQL, Linux, Docker, Jenkins] **July 2022 – August 2025**

- Accelerated multi-investment, multi-currency report generation via parallelism in C++, slashing processing time by 99.8%.
- Diagnosed performance issues in portfolio and asset reports, enhancing throughput and scalability.
- Overhauled database schema for normalization, removing redundancy and quadrupling reporting speed.
- Revamped dashboard with RSL scripting, boosting search visibility and UX by 30%.
- Implemented a dropdown asset-selection feature in C++/RSL, increasing client engagement by 25%.
- Developed Python–MySQL functionality to exclude closed portfolios from the GMM model, reducing processing errors.
- Engineered a Python diagnostic script to validate environment configurations, ensuring reliable development setups.
- Crafted a Python comparison tool using dictionaries to detect and log discrepancies across multiple runs.

Software Engineer –LTI, India [Hyperion Infra, Oracle EPM, Java, MySQL, HTML, CSS, JavaScript] **Aug 2018 – Nov, 2020**

- Orchestrated ETL migration across UAT, DEV, and QA using Informatica PowerCenter, streamlining data workflows.
- Led OBIEE repository migrations with Oracle BI Admin, Enterprise Manager, and Data Integrator.
- Developed an interactive website (HTML/CSS/JS) enabling users to query bus schedules, select seats, & view past bookings.
- Engineered a Java-based banking system with OOP inheritance for bank accounts, deposits and withdrawals.
- Outlined Online Banking Test Driven Model boiler plate using Java and performed Junit Testing.

ACADEMIC PROJECTS

Building & Performance Testing of a Highly Available Website using GCP Serverless Computing [JavaScript, Python, GCP]

- Deployed a website to track International Space Station and display location on Google Maps using JavaScript by utilizing features such as google cloud function, google cloud SQL and CRON job scheduler.
- Analyzed website performance by tracking the number of requests, latency and container count using Apache Benchmark tool.

Containers & Mini Docker [Python, Docker, Sysbench, QEMU, GCP]

- Implemented a Docker like tool called Mini Docker in Python to create container instances with namespace & control groups.
- Evaluated the performance of Docker container and QEMU using sysbench tool by running CPU and File I/O test modes.

Smart Home Automation using Alexa [Python, flask, AWS, XML, JSON]

- Instituted functionalities such as turning devices ON/OFF, adjusting brightness, setting schedules, creating and controlling device zones and providing security.
- Trained Alexa with intents and utterances in JSON using AWS Lambda function, maintained and tracked the database in xml.

Sales Forecasting using Time Series Data Analysis [Python]

- Forecasted sales of an e-commerce business by building an ARIMA model and calculated the model's accuracy using RMSE.

Mini Shell [C]

- Developed a shell (console), including system calls like fork, exec, wait, pipe to let the user run the commands with arguments, input and output file redirection and multiple commands separated by pipes.

EDUCATION

Master of Science, Computer Science State @ University of New York at Binghamton, New York

Feb 2021 – May 2022

Bachelor of Engineering, Information Technology @ University of Pune, India

Aug 2014 – May 2018

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, C, JavaScript

Web Technologies: HTML5, CSS3, PHP, jQuery

Databases: MySQL, MongoDB, Cassandra

Cloud & Containers: GCP, Docker, AWS (Lambda Function)

Version Control: GitHub

ACHIEVEMENTS & PUBLISHPMENT

- Presented and published research paper titled “[*Smart Homes Using Alexa and Power Line Communication in IoT | SpringerLink*](#)” at an annual ICCCNT conference in Springer.

April, 2018